MASTER LIST — Java + Spring Boot Interview Questions (0–2 YOE)

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Here's a curated master list of **real Java + Spring Boot interview questions** repeatedly asked to **0–2 years experience candidates/freshers** in both service-based and product-based companies.

SECTION 1 — Core Java Questions

OOPS

1. Explain all OOP principles with real examples.
2. Difference between abstraction and encapsulation.
3. Difference between abstract class and interface.
4. Can interface have methods with body?
5. What is multiple inheritance and how Java handles it?
6. Difference between method overloading and overriding.
7. What is runtime polymorphism?
8. Why is Java platform independent?
9. Explain JVM, JDK and JRE.
10. Explain memory allocation in Java.

String Questions

11. Difference between String, StringBuilder and StringBuffer.
12. Why are Strings immutable?
13. What is String Constant Pool?
14. == vs equals().
15. How many objects are created in:

String s = new String("hello");

16. Reverse a string without using built-in methods.
17. Find duplicate characters in a string.

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18. Check if string is palindrome.
19. Find first non-repeated character.

Collections Framework

20. Difference between ArrayList and LinkedList.
21. Difference between HashMap and Hashtable.
22. Internal working of HashMap.
23. What is hashing?
24. What is collision in HashMap?
25. Why HashMap allows one null key?
26. Difference between HashSet and TreeSet.
27. Difference between fail-fast and fail-safe iterator.
28. What is ConcurrentModificationException?
29. Difference between Comparable and Comparator.
30. Sort Employee objects based on salary/name.
31. What is priority queue?
32. Difference between Queue and Deque.
33. Time complexities of HashMap operations.
34. How does HashMap resize internally?
35. Difference between List and Set.

Exception Handling

36. Difference between checked and unchecked exception.
37. Difference between throw and throws.
38. What is finally block?
39. Can finally block skip execution?
40. Create custom exception.
41. Difference between final, finally and finalize.
42. What is try-with-resources?
43. Exception hierarchy in Java.

Multithreading

44. Difference between process and thread.
45. Difference between Runnable and Thread class.
46. What is synchronization?
47. What is deadlock?

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48. Difference between wait(), notify(), notifyAll().
49. What is volatile keyword?
50. What is thread pool?
51. What is ExecutorService?
52. Difference between sleep() and wait().
53. What is race condition?
54. Explain CountDownLatch.
55. What is CompletableFuture?
56. Explain producer-consumer problem.

Java 8 Questions

57. What are Java 8 features?
58. What is Lambda expression?
59. Functional interfaces examples.
60. What is Stream API?
61. Difference between map() and flatMap().
62. Difference between filter() and map().
63. What is method reference?
64. What is Optional?
65. Find duplicate elements using streams.
66. Find second highest salary using streams.
67. Group employees by department.
68. Difference between intermediate and terminal operations.
69. Parallel stream vs normal stream.
70. Collectors methods in Java 8.
71. Sort list using streams.
72. Convert list to map using streams.

Coding Questions Frequently Asked

73. Reverse linked list.
74. Find duplicates in array.
75. Two sum problem.
76. Palindrome number/string.
77. Binary search implementation.
78. LRU cache implementation.
79. Longest substring without repeating characters.
80. Fibonacci using recursion.
81. Factorial program.
82. Check anagram.

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83. Merge two sorted arrays.
84. Sort hashmap by value.
85. Find missing number in array.
86. Remove duplicates from list.
87. Detect cycle in linked list.
88. Build tree from preorder + inorder. (LeetCode)
89. Longest valid parentheses. (LeetCode)

SECTION 2 — Spring Framework Questions

90. What is Spring Framework?
91. What is Dependency Injection?
92. Types of DI.
93. What is IOC container?
94. Bean lifecycle in Spring.
95. Difference between BeanFactory and ApplicationContext.
96. What are Spring modules?
97. What is @Autowired?
98. Constructor vs Setter Injection.
99. What is stereotype annotation?
100. Difference between @Component, @Service, @Repository, @Controller.
101. What is component scanning?
102. What is @Qualifier?
103. What is Spring AOP?
104. What are aspects, join points and advice?
105. What is transaction management?
106. What is @Transactional?

SECTION 3 — Spring Boot Questions

107. What is Spring Boot?
108. Advantages of Spring Boot.
109. What is @SpringBootApplication?
110. Explain auto-configuration.
111. What are starter dependencies?

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112. Difference between Spring and Spring Boot.
113. Explain embedded server.
114. What is Spring Boot Actuator?
115. What is Spring Boot DevTools?
116. What is application.properties?
117. application.properties vs application.yml.
118. What is profile in Spring Boot?
119. Explain @Configuration.
120. Difference between @RestController and @Controller.
121. What is ResponseEntity?
122. Explain REST APIs.
123. Difference between PUT and POST.
124. Difference between PATCH and PUT.
125. Explain HTTP status codes.
126. What is DTO?
127. How do you validate request body?
128. What is @Valid?
129. Explain exception handling in Spring Boot.
130. What is @ControllerAdvice?
131. Explain global exception handling.
132. What is Swagger/OpenAPI? (GeeksforGeeks)
133. How to connect DB in Spring Boot? (GeeksforGeeks)
134. What is Spring Data JPA?
135. Difference between CrudRepository and JpaRepository.
136. What is Hibernate?
137. Explain lazy vs eager loading.
138. What is N+1 problem?
139. Difference between save() and saveAndFlush().
140. What is pagination?
141. What is caching?
142. Explain microservices basics.
143. Monolith vs microservices.
144. What is API Gateway?
145. What is Eureka Server?
146. What is Feign Client?
147. What is Circuit Breaker?
148. Explain distributed tracing.
149. Explain Spring Security basics.
150. JWT authentication flow.
151. What is BCryptPasswordEncoder?
152. Difference between authentication and authorization.
153. What is CORS?
154. Explain CSRF.
155. How do you secure REST APIs?

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SECTION 4 — Database + SQL Questions

156. Difference between SQL and NoSQL.
157. Explain normalization.
158. What is indexing?
159. Primary key vs unique key.
160. Foreign key.
161. Difference between DELETE, DROP and TRUNCATE.
162. What is JOIN?
163. Types of joins.
164. Difference between WHERE and HAVING.
165. What is ACID property?
166. What is transaction?
167. Write query to find second highest salary.
168. Difference between clustered and non-clustered index.
169. What is database sharding?
170. Explain optimistic and pessimistic locking.

SECTION 5 — System Design + Backend Basics

171. Design URL shortener.
172. Design parking system.
173. Design vehicle rental system.
174. Explain caching strategies.
175. Explain load balancer.
176. What is Kafka?
177. What is Redis?
178. Explain rate limiting.
179. How to scale backend APIs?
180. Monolithic vs distributed architecture.
181. Explain API versioning.
182. What is idempotency?
183. Explain SOLID principles.
184. Explain Dependency Inversion Principle.
185. Explain Singleton design pattern.

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186. Thread-safe singleton implementation.

SECTION 6 — Frequently Asked HR + Resume Questions

187. Explain your project architecture.
188. Explain your role in project.
189. Biggest challenge in project?
190. Explain one bug you fixed.
191. Why Spring Boot?
192. Why should we hire you?
193. Explain your internship/project deeply.
194. Explain API flow in your project.
195. How do frontend and backend communicate?
196. Explain deployment process.
197. Explain CI/CD basics.
198. Explain Git workflow.
199. Difference between merge and rebase.
200. Explain your favorite project end-to-end.

Companies Where These Questions Were Repeatedly Reported

Service-Based Companies

- Infosys
- Tata Consultancy Services
- Wipro
- Accenture
- Capgemini
- Cognizant
- Deloitte
- Tech Mahindra
- HCLTech

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Product-Based Companies

- Swiggy
- Rakuten
- Apple
- Amazon
- Flipkart
- Paytm
- Oracle
- Zoho

Most Repeated Topics Across 2024–2026 Interviews

1. Internal working of HashMap
2. Java 8 Streams
3. REST APIs
4. Spring Boot annotations
5. Multithreading basics
6. SQL joins + indexing
7. Exception handling
8. Collections framework
9. OOPS concepts
10. Microservices basics
11. JWT authentication
12. Project explanation
13. LLD basics
14. SOLID principles
15. Coding using streams

High-Value Observation From Real Interview Trends

For **0–2 YOE**, companies are heavily focusing on:

- Java fundamentals
- Spring Boot annotations/configuration
- REST APIs
- SQL
- Java 8 streams
- Collections internals

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- Project explanation
- Basic LLD
- Coding round with easy-medium DSA

Deep system design is usually NOT expected for freshers, but API flow and backend understanding are expected increasingly in 2025–2026.

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